

order to compute the TED score. Inference timing results for all experiments were obtained from the same machine on a single core with AMD EPYC 7763 CPU @2.45 GHz.

5.1 Hyper Parameter Optimization

We have chosen the PubTabNet data set to perform HPO, since it includes a highly diverse set of tables. Also we report TED scores separately for simple and complex tables (tables with cell spans). Results are presented in Table. 1. It is evident that with OTSL, our model achieves the same TED score and slightly better mAP scores in comparison to HTML. However OTSL yields a *2x speed up* in the inference runtime over HTML.

Table 1. HPO performed in OTSL and HTML representation on the same transformer-based TableFormer [9] architecture, trained only on PubTabNet [22]. Effects of reducing the # of layers in encoder and decoder stages of the model show that smaller models trained on OTSL perform better, especially in recognizing complex table structures, and maintain a much higher mAP score than the HTML counterpart.

# enc-layers	# dec-layers	Language	TEDs			mAP (0.75)	Inference time (secs)
			simple	complex	all		
6	6	OTSL	0.965	0.934	0.955	0.88	2.73
		HTML	0.969	0.927	0.955	0.857	5.39
4	4	OTSL	0.938	0.904	0.927	0.853	1.97
		HTML	0.952	0.909	0.938	0.843	3.77
2	4	OTSL	0.923	0.897	0.915	0.859	1.91
		HTML	0.945	0.901	0.931	0.834	3.81
4	2	OTSL	0.952	0.92	0.942	0.857	1.22
		HTML	0.944	0.903	0.931	0.824	2

5.2 Quantitative Results

We picked the model parameter configuration that produced the best prediction quality (enc=6, dec=6, heads=8) with PubTabNet alone, then independently trained and evaluated it on three publicly available data sets: PubTabNet (395k samples), FinTabNet (113k samples) and PubTables-1M (about 1M samples). Performance results are presented in Table. 2. It is clearly evident that the model trained on OTSL outperforms HTML across the board, keeping high TEDs and mAP scores even on difficult financial tables (FinTabNet) that contain sparse and large tables.

Additionally, the results show that OTSL has an advantage over HTML when applied on a bigger data set like PubTables-1M and achieves significantly improved scores. Finally, OTSL achieves faster inference due to fewer decoding steps which is a result of the reduced sequence representation.

Table 2. TSR and cell detection results compared between OTSL and HTML on the PubTabNet [22], FinTabNet [21] and PubTables-1M [14] data sets using TableFormer [9] (with enc=6, dec=6, heads=8).

Data set	Language	TEDs			mAP(0.75)	Inference time (secs)
		simple	complex	all		
PubTabNet	OTSL	0.965	0.934	0.955	0.88	2.73
	HTML	0.969	0.927	0.955	0.857	5.39
FinTabNet	OTSL	0.955	0.961	0.959	0.862	1.85
	HTML	0.917	0.922	0.92	0.722	3.26
PubTables-1M	OTSL	0.987	0.964	0.977	0.896	1.79
	HTML	0.983	0.944	0.966	0.889	3.26

5.3 Qualitative Results

To illustrate the qualitative differences between OTSL and HTML, Figure 5 demonstrates less overlap and more accurate bounding boxes with OTSL. In Figure 6, OTSL proves to be more effective in handling tables with longer token sequences, resulting in even more precise structure prediction and bounding boxes.

Fig. 5. The OTSL model produces more accurate bounding boxes with less overlap (E) than the HTML model (D), when predicting the structure of a sparse table (A), at twice the inference speed because of shorter sequence length (B),(C). "PMC2807444_006_00.png" PubTabNet.

